

SUPPLY CHAIN MANAGEMENT IN ERP

List of Topic to be Discuss in this Lecture

- ✓ Introduction to Supply Chain Management (SCM)
- ✓ Key Concepts in Supply Chain Management
- ✓ SCM in ERP Systems
- ✓ Key Features of ERP for SCM
- ✓ Benefits of SCM in ERP
- ✓ Challenges in Implementing ERP for SCM
- ✓ ERP Supply Chain Management Workflow / Business Process
- ✓ Case Study

Introduction to Supply Chain Management (SCM)

- **Definition:** Supply Chain Management involves planning, controlling, and executing the flow of goods, information, and finances from raw material sourcing to delivering finished goods to customers.
- **Objective:** To optimize efficiency, reduce costs, and enhance customer satisfaction by managing all supply chain activities cohesively.
- **Role in ERP:** ERP systems integrate SCM functions, providing real-time data, seamless collaboration, and enhanced visibility across the supply chain.

Key Concepts in Supply Chain Management

1. **Supply Chain Components:**
 - **Suppliers:** Provide raw materials or components.
 - **Manufacturers:** Convert raw materials into finished goods.
 - **Distributors:** Store and transport goods to various locations.
 - **Retailers:** Sell goods to end consumers.
 - **Customers:** Ultimate recipients of goods or services.
2. **SCM Processes:**
 - **Planning:** Demand forecasting, production scheduling, and inventory planning.
 - **Procurement:** Sourcing materials and managing supplier relationships.

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- **Manufacturing:** Converting raw materials into products.
 - **Logistics:** Warehousing, transportation, and delivery.
 - **Returns Management:** Handling defective goods or post-sale returns.
3. **Flows in SCM:**
- **Material Flow:** Physical movement of goods.
 - **Information Flow:** Communication and data sharing.
 - **Financial Flow:** Payment transactions and credit terms.

SCM in ERP Systems

- **Overview:** ERP systems streamline SCM by integrating all related processes into a unified platform, allowing real-time monitoring, analytics, and decision-making.

Key Features of ERP for SCM

1. **End-to-End Visibility:**
 - Real-time tracking of goods and materials across the supply chain.
 - Enables proactive responses to disruptions.
2. **Demand Forecasting:**
 - Uses historical data and market trends to predict future demand.
 - Prevents overproduction or understocking.
3. **Inventory Optimization:**
 - Balances stock levels with demand and supply dynamics.
 - Reduces carrying costs and stockouts.
4. **Collaboration Tools:**
 - Enhances communication between suppliers, manufacturers, and distributors.
 - Encourages joint problem-solving and planning.
5. **Automation and Workflow Management:**
 - Automates routine tasks like order processing and replenishment.
 - Streamlines approvals and reduces manual errors.
6. **Analytics and Reporting:**
 - Provides insights into supplier performance, order trends, and logistics costs.
 - Facilitates data-driven decision-making.

Benefits of SCM in ERP

1. **Operational Efficiency:**
 - Minimizes delays and bottlenecks.
 - Enhances resource utilization.
2. **Cost Savings:**
 - Optimized procurement and inventory reduce costs.
 - Efficient logistics lower transportation expenses.
3. **Improved Customer Service:**
 - Faster order fulfillment and reliable deliveries enhance customer satisfaction.

- Transparent order tracking boosts trust.
- 4. **Risk Mitigation:**
 - Identifies potential supply chain disruptions.
 - Implements contingency plans proactively.
- 5. **Compliance and Sustainability:**
 - Ensures adherence to industry standards and environmental regulations.
 - Tracks carbon footprints and waste reduction efforts.

Challenges in Implementing ERP for SCM

1. **Integration Complexity:**
 - Difficulty in aligning ERP with existing systems or processes.
2. **High Costs:**
 - Significant investment required for implementation and training.
3. **Change Management:**
 - Resistance from employees or partners to adopt new systems.
4. **Data Accuracy:**
 - Errors in data entry can lead to inaccurate decisions.

ERP Supply Chain Management Workflow / Business Process

1. **Demand Forecasting:**
 - ERP predicts demand using historical data.
2. **Procurement Planning:**
 - System generates purchase requisitions based on inventory levels.
3. **Production Scheduling:**
 - Aligns manufacturing timelines with demand forecasts.
4. **Inventory Management:**
 - Updates stock levels in real-time after production or receipt of goods.
5. **Order Fulfillment:**
 - Tracks orders from placement to delivery.
6. **Performance Monitoring:**
 - Monitors KPIs like order accuracy, lead times, and cost per shipment.

Case Study

Below is a detailed step-by-step example of how Supply Chain Management (SCM) works in an ERP system, integrating various processes to ensure seamless operations.

Scenario

Company Type: A mid-sized manufacturing company producing consumer electronics.
Objective: Fulfill a bulk order of 5,000 units of a new product while minimizing costs and meeting the delivery deadline.

Workflow Steps

1. Demand Forecasting

- The ERP system uses historical sales data and market trends to predict demand.
- A forecast shows the need to produce 5,000 units over the next month to fulfill the order.
- The system flags any potential shortfall in inventory.

2. Procurement Planning

- Based on the forecast, the Material Management (MM) module analyzes raw material requirements.
- The system generates purchase requisitions for items not available in the inventory (e.g., microchips, casings, and wiring).
- Supplier performance and past delivery reliability are evaluated by the Supplier Relationship Management (SRM) module.
- Purchase orders (POs) are automatically created and sent to selected suppliers.

3. Production Planning

- The Production Planning (PP) module creates a manufacturing schedule aligning with the delivery deadline.
- The system allocates resources like machinery and workforce.
- A bill of materials (BOM) is generated for the production process, ensuring all components are accounted for.

4. Inbound Logistics

- The ERP system tracks raw material shipments from suppliers using the Transportation Management (TM) module.
- As materials arrive, the Warehouse Management (WM) module records goods receipt and updates inventory levels in real-time.

5. Manufacturing Execution

- The production team follows the schedule and BOM provided by the PP module.
- The ERP system monitors production progress, highlighting any delays or issues.
- Quality control checks are conducted, and results are logged in the system.

6. Inventory Management

- Finished goods are stored in the warehouse.
- The ERP system updates inventory records, ensuring accurate stock levels are visible for order fulfillment.
- Safety stock levels are maintained for unplanned demands.

7. Order Fulfillment

- The Sales and Distribution (SD) module processes the bulk order from the customer.

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- The system automatically picks, packs, and schedules the shipment of 5,000 units.
- Real-time tracking information is shared with the customer through a portal or notification system.

8. Outbound Logistics

- The Transportation Management (TM) module plans optimal shipping routes and modes.
- The system ensures on-time delivery, minimizing costs by consolidating shipments where possible.
- Delivery confirmation is logged in the ERP system.

9. Invoice and Payment Processing

- The Financial Accounting (FI) module generates an invoice based on the sales order.
- The customer processes the payment, and the ERP system reconciles it with the invoice.
- The payment is recorded, and financial reports are updated.

10. Performance Monitoring and Reporting

- The ERP system generates post-operation analytics, including:
 - Supplier delivery performance.
 - Production efficiency and downtime.
 - Logistics costs and delivery accuracy.
- Insights from these reports are used to optimize future supply chain processes.

Key ERP Modules Involved

- **Material Management (MM):** Procurement and inventory tracking.
- **Production Planning (PP):** Scheduling and resource allocation.
- **Warehouse Management (WM):** Stock handling and storage optimization.
- **Transportation Management (TM):** Logistics and shipment planning.
- **Sales and Distribution (SD):** Order processing and customer interaction.
- **Financial Accounting (FI):** Payment and financial reporting.

Outcome

- **On-time Delivery:** 5,000 units delivered within the agreed timeframe.
- **Cost Savings:** Optimized procurement and logistics reduced costs by 15%.
- **Customer Satisfaction:** Accurate and timely communication ensured a high level of customer satisfaction.
- ✓ **Continuous Improvement:** Performance data highlighted areas for further supply chain enhancements.

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Steps in the Procurement Planning Phase to Avoid Delays:

should take the following steps:

- **Identify Alternative Suppliers:** Use the ERP's Supplier Relationship Management (SRM) module to search for reliable suppliers with available stock of microchips.
- **Send Purchase Orders (POs) Quickly:** The Material Management (MM) module can automatically generate and send POs to the selected suppliers to minimize delays.
- **Negotiate Expedited Delivery:** Engage with suppliers to ensure priority shipping for critical components to meet the production schedule.

Role of the SRM Module in Mitigating Supply Chain Risk:

The SRM module helps by:

- **Evaluating Supplier Performance:** It provides insights into past supplier reliability, lead times, and quality.
- **Identifying Backup Suppliers:** The system suggests alternative suppliers based on their performance records.
- **Managing Contracts and Relationships:** It allows TechWare Inc. to maintain strong partnerships with key suppliers and negotiate better terms during emergencies.